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Interpretation of molecular tests for encapsulated bacteria

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14/09/2025

Setting up a PCR

1) Lab design

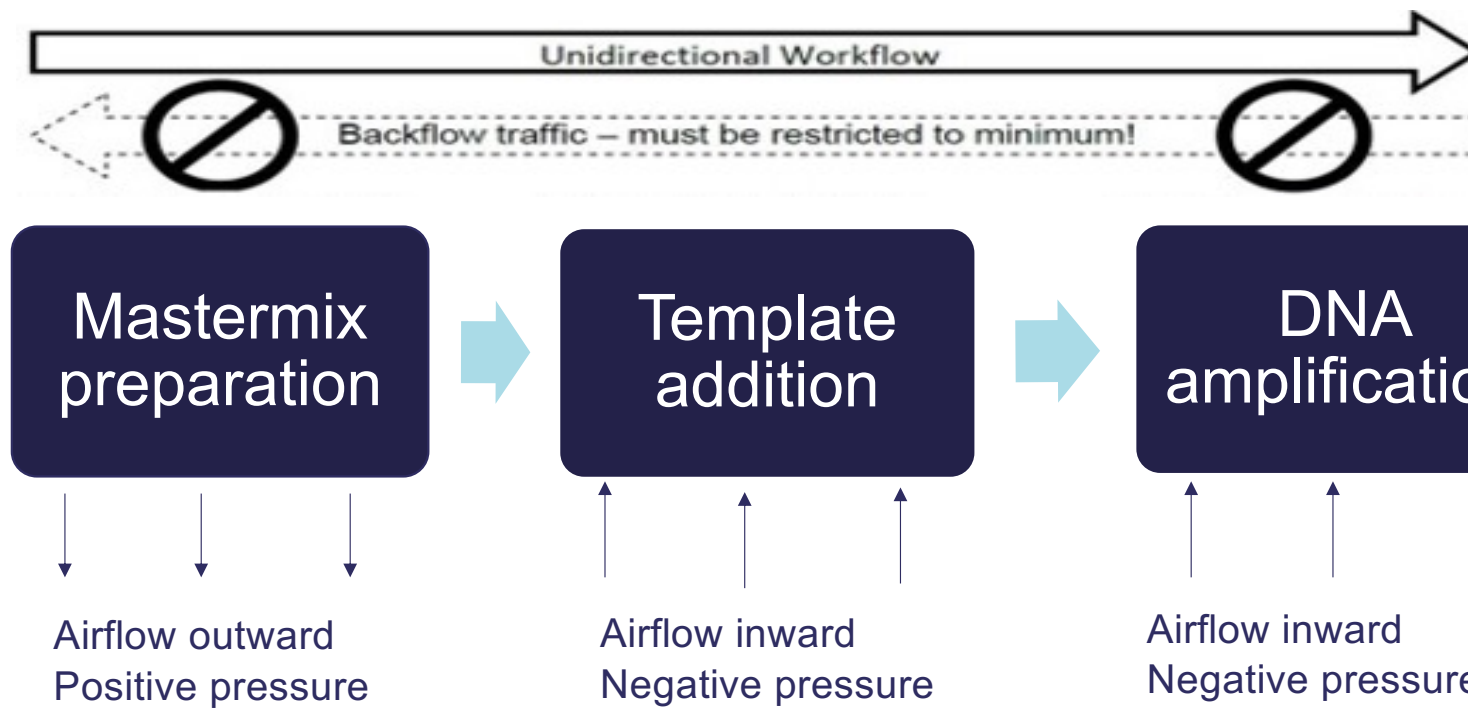
- Ideally, a PCR lab should have 3 rooms each designed for specific tasks
- The first room should be exclusively used for Master mix preparation
- The second room should be dedicated for template addition,
- The third room should be dedicated for DNA amplification
- follows a unidirectional workflow. Use dedicated equipment for each area, e.g. 2 different sets of pipettes.



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Setting up a PCR

2) Equipment

- Laminar flow or biosafety cabinet
- Pipettes
- Thermocycler





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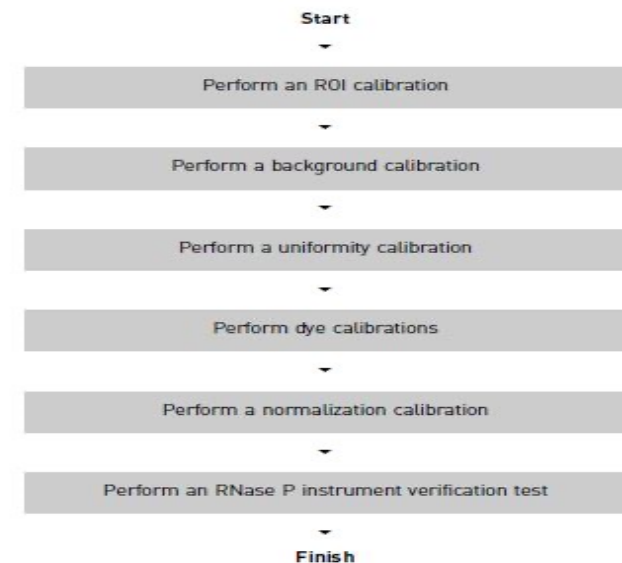
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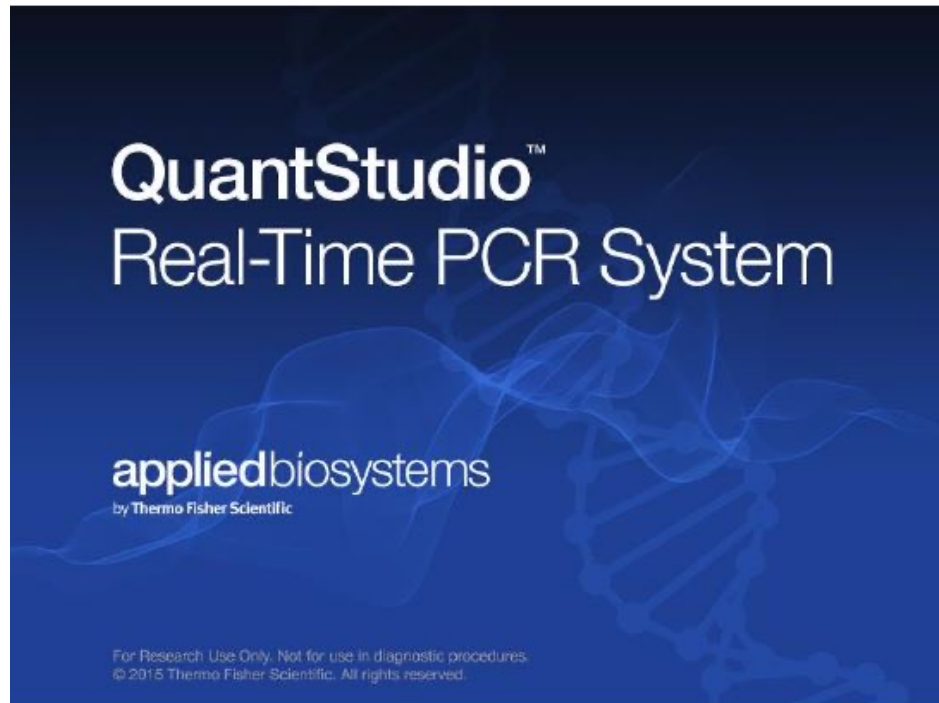
Setting up a PCR

Thermocycler

- Set up your run on a calibrated instrument
- To ensure optimal performance of the thermal cycler or Sequence Detection System, it is strongly recommended that the thermal cycler is calibrated regularly. Calibration varies by instrument.



The Quantsudio 7 flex Equipment



- Opens files from both Quantsudio and Viia7 machine
- No license required for download and analysis

<https://www.thermofisher.com/gm/en/home/global/forms/life-science/quantstudio-6-7-flex-software.html>



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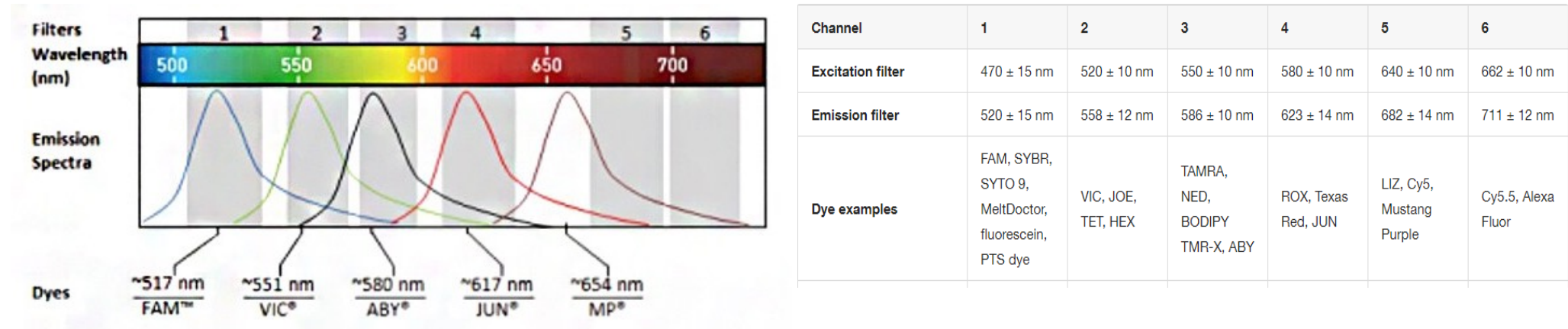


Types of real time PCR reaction based on number of gene targets

Monoplex: only one gene is amplified in each PCR reaction well.

Multiplex: two or more target genes are amplified in the same reaction, using the same reagent mix.

For multiplex reactions, choose dyes with little to no overlap in their emission spectra.



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How do you determine the dye to use for your assay?

1. Choose dyes that are compatible with your instrument
2. Avoid overlap of emission spectra. Choose dyes with appropriate qPCR dye spectrum with appropriate excitation wavelengths and little to no overlap in their emission spectra
3. Choose dyes that can be calibrated on your instrument
4. Consider cost and availability



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qPCR assays for meningitis detection

- 1) **Monoplex** detection of Spn, Nm and Hi
- 2) **Multiplex** detection of Spn Nm and Hi
- 3) **Monoplex** for serogrouping/typing of Nm and Hi
- 4) **Multiplex** serotyping of Spn



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Setting up a PCR

3) PCR controls

The main kinds of controls used in PCR applications are:

- Negative and positive PCR controls.
- Negative and positive DNA extraction controls
- Internal PCR assay controls



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Negative PCR controls

Negative PCR Control	Sample PCR result	Inferences and next steps
Not included	Amplicons from samples observed	
Included and negative	Amplicons from samples observed	
Included and negative	No amplicons observed from samples	
Included and positive	Amplicons either observed or not observed	

Negative PCR controls

Negative PCR Control	Sample PCR result	Inferences and next steps
Not included	Amplicons from samples observed	The PCR worked but might be contaminated. Repeat the experiment with a negative control.
Included and negative	Amplicons from samples observed	
Included and negative	No amplicons observed from samples	
Included and positive	Amplicons either observed or not observed	

Negative PCR controls

Negative PCR Control	Sample PCR result	Inferences and next steps
Not included	Amplicons from samples observed	The PCR worked but might be contaminated. Repeat the experiment with a negative control.
Included and negative	Amplicons from samples observed	The PCR worked and is unlikely to be contaminated.
Included and negative	No amplicons observed from samples	
Included and positive	Amplicons either observed or not observed	

Negative PCR controls

Negative PCR Control	Sample PCR result	Inferences and next steps
Not included	Amplicons from samples observed	The PCR worked but might be contaminated. Repeat the experiment with a negative control.
Included and negative	Amplicons from samples observed	The PCR worked and is unlikely to be contaminated.
Included and negative	No amplicons observed from samples	The PCR may have worked (if testing for presence or absence), or may have failed (if a result was expected). Repeat the experiment with a positive PCR control to confirm for a genuine absence of the target, and if an amplicon was expected then also use a positive DNA extraction control to distinguish between PCR failure and DNA extraction failure.
Included and positive	Amplicons either observed or not observed	

Negative PCR controls

Negative PCR Control	Sample PCR result	Inferences and next steps
Amplicons from samples observed	Not included	The PCR worked but might be contaminated. Repeat the experiment with a negative control.
Amplicons from samples observed	Included and negative	The PCR worked and is unlikely to be contaminated.
No amplicons observed from samples	Included and negative	The PCR may have worked (if testing for presence or absence), or may have failed (if a result was expected). Repeat the experiment with a positive PCR control to confirm for a genuine absence of the target, and if an amplicon was expected then also use a positive DNA extraction control to distinguish between PCR failure and DNA extraction failure.
Amplicons either observed or not observed	Included and positive	The PCR worked but has some form of systemic contamination It may be difficult to distinguish successful PCR products from PCR contaminants.

Positive and negative controls

Negative PCR control	Positive PCR control	Sample PCR result	Inferences and next steps
Negative	Positive	Amplicons observed	
Negative	Negative	Amplicons observed	
Positive	Positive	Amplicons observed	
Negative	Positive	No amplicons observed (If amplifications are expected)	

Positive and negative controls

Negative PCR control	Positive PCR control	Sample PCR result	Inferences and next steps
Negative	Positive	Amplicons observed	The PCR worked in general and for your samples, and is unlikely to be contaminated
Negative	Negative	Amplicons observed	The PCR did not work in general, and you should troubleshoot the reaction. If you observe amplicons in the positive control, the reaction components are likely good, and the problem is likely with the template DNA. If you do not observe amplicons in the positive control, the reaction components may be the problem.
Positive	Positive	Amplicons observed	
Negative	Positive	No amplicons observed (If amplifications are expected)	

Positive and negative controls

Negative PCR control	Positive PCR control	Sample PCR result	Inferences and next steps
Negative	Positive	Amplicons observed	The PCR worked in general and for your samples, and is unlikely to be contaminated
Negative	Negative	Amplicons observed	The PCR worked in general and for your samples, and is unlikely to be contaminated. Your positive control needs troubleshooting or replacing.
Positive	Positive	Amplicons observed	
Negative	Positive	No amplicons observed (If amplifications are expected)	

Positive and negative controls

Negative PCR control	Positive PCR control	Sample PCR result	Inferences and next steps
Negative	Positive	Amplicons observed	The PCR worked in general and for your samples, and is unlikely to be contaminated
Negative	Negative	Amplicons observed	The PCR worked in general and for your samples, and is unlikely to be contaminated. Your positive control needs troubleshooting or replacing.
Positive	Positive	Amplicons observed	The PCR in general worked and there is systemic contamination present. It may be difficult or impossible to distinguish successful sample PCRs from contamination.
Negative	Positive	No amplicons observed (If amplifications are expected)	

Positive and negative controls

Negative PCR control	Positive PCR control	Sample PCR result	Inferences and next steps
Negative	Positive	Amplicons observed	The PCR worked in general and for your samples, and is unlikely to be contaminated
Negative	Negative	Amplicons observed	The PCR worked in general and for your samples, and is unlikely to be contaminated. Your positive control needs troubleshooting or replacing.
Positive	Positive	Amplicons observed	The PCR in general worked and there is systemic contamination present. It may be difficult or impossible to distinguish successful sample PCRs from contamination.
Negative	Positive	No amplicons observed (If amplifications are expected)	The PCR in general worked and your samples failed. Troubleshoot your DNA extractions.

Real time PCR data analysis

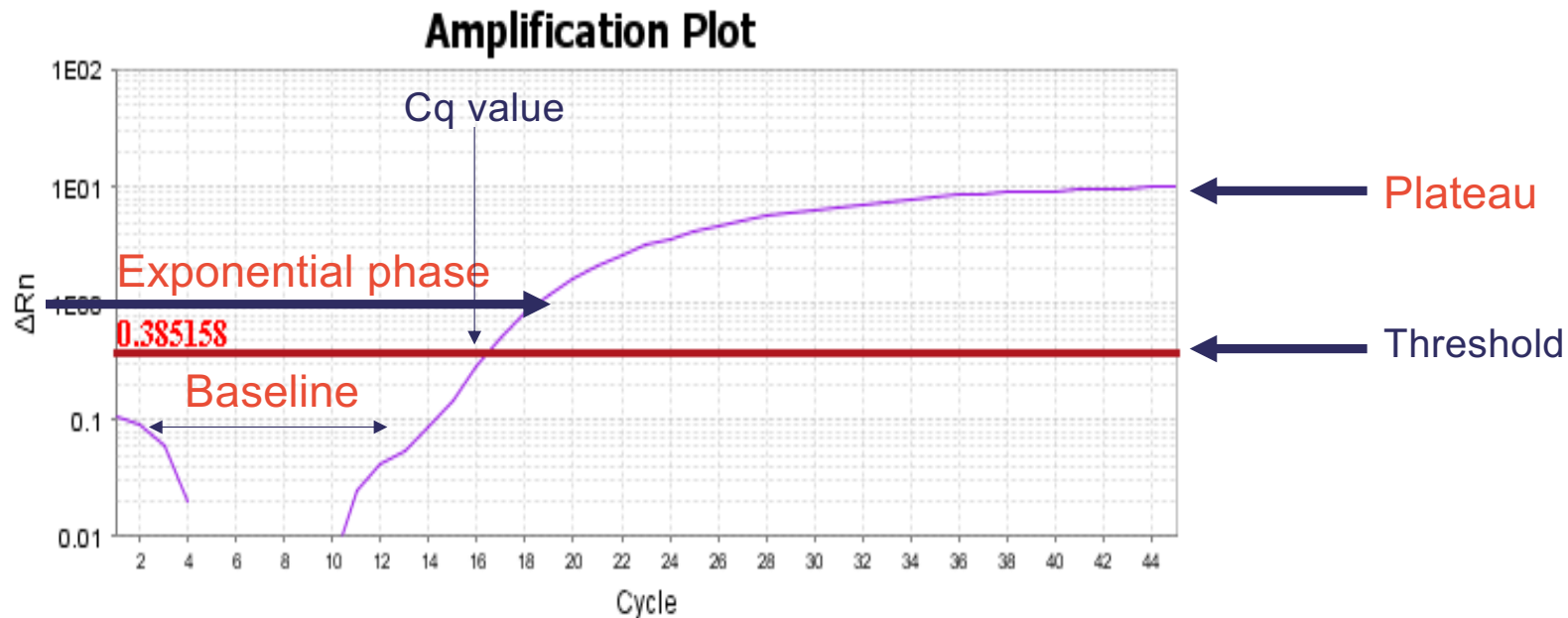


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Amplification curve



Threshold: the threshold definition refers to the point at which the fluorescence signal generated during the PCR reaction crosses a predetermined threshold value.

Cq value : the cycle number at which the fluorescent signal of a sample's DNA amplification crosses a defined threshold,.



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Review and analyse results

Steps to follow

Step 1: Check assay controls

- Positive control
- Negative control

Step 2: Analyse data

Step 3: Save changes and export data



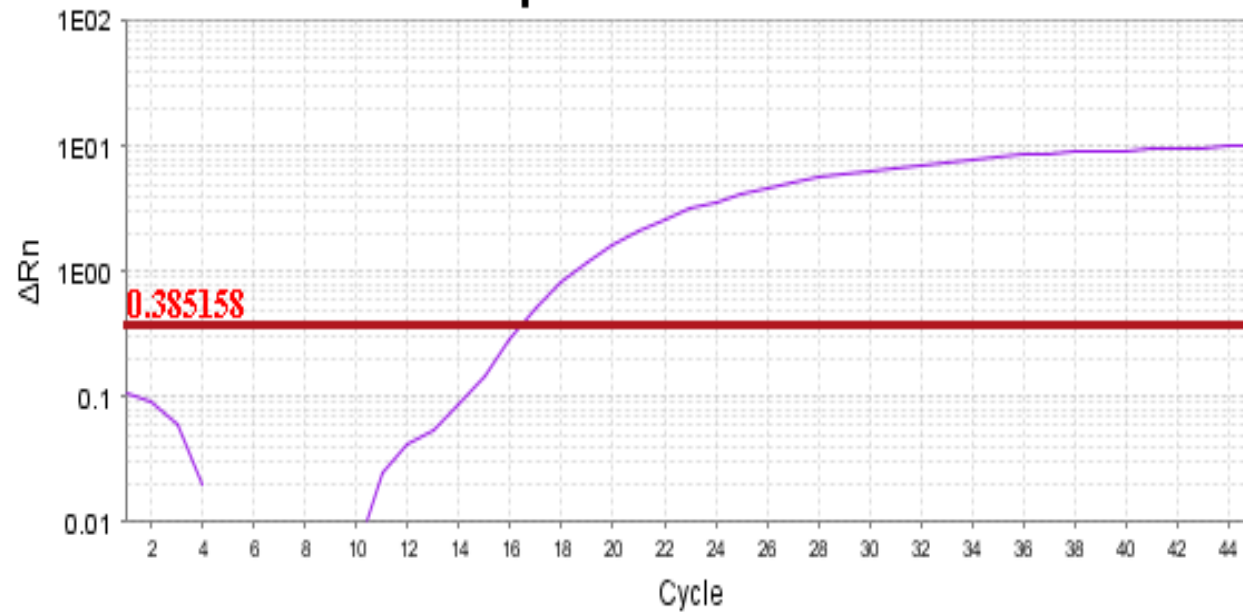
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Positive sample

Amplification Plot



Baseline	Baseline Start	Baseline End
AUTO	3	12

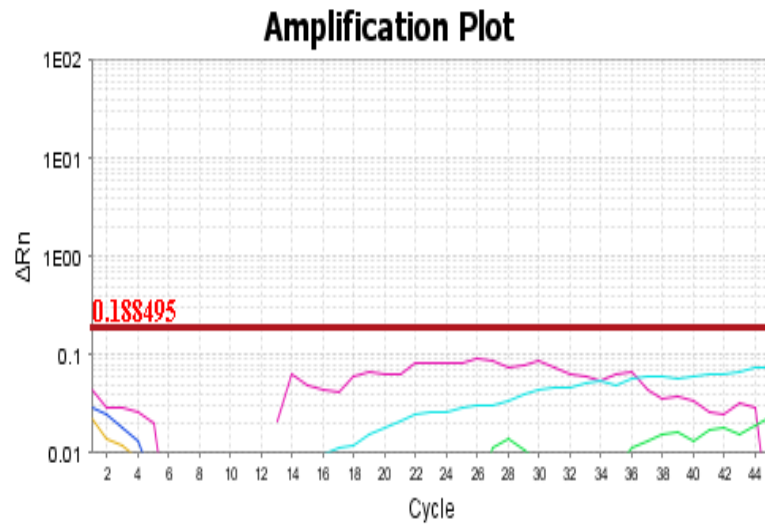


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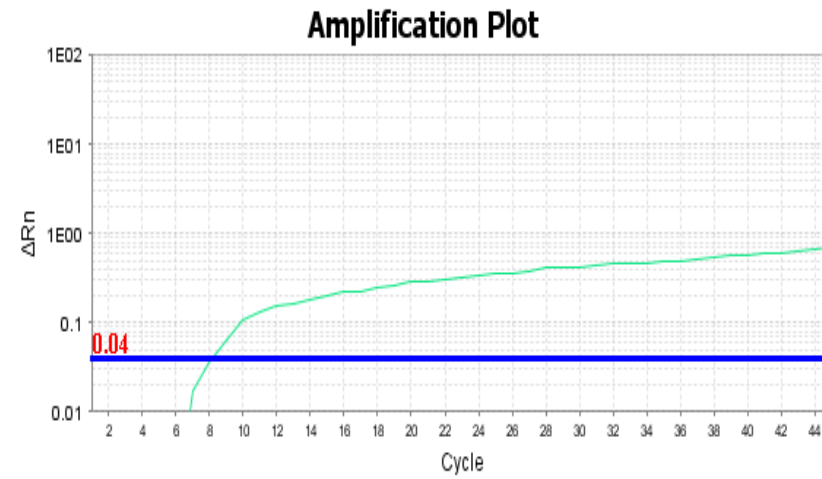
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Negative samples



Baseline	Baseline Start	Baseline End
AUTO	3	44



Baseline	Baseline Start	Baseline End
AUTO	3	6

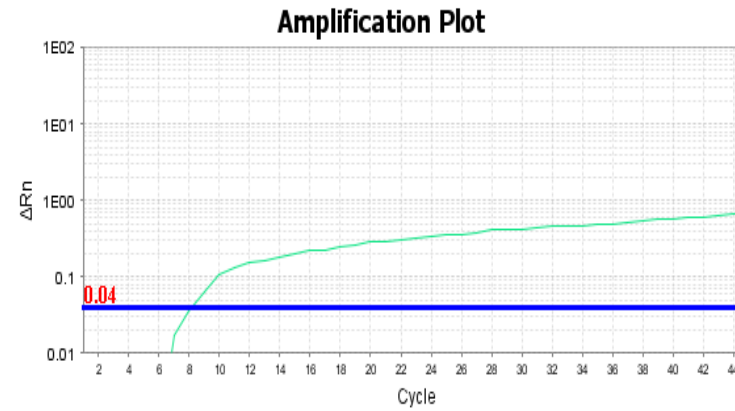
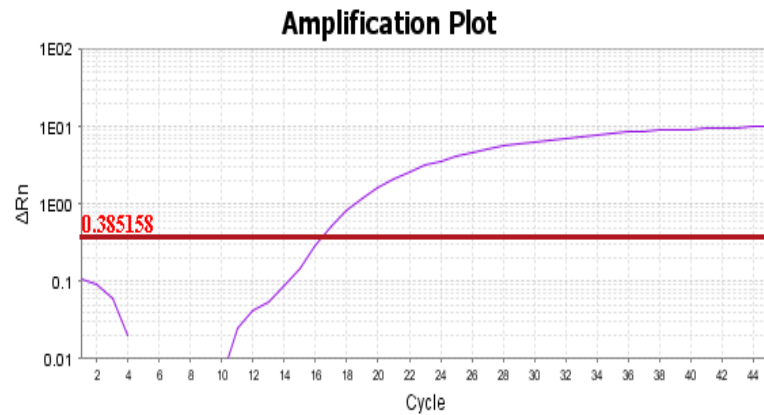


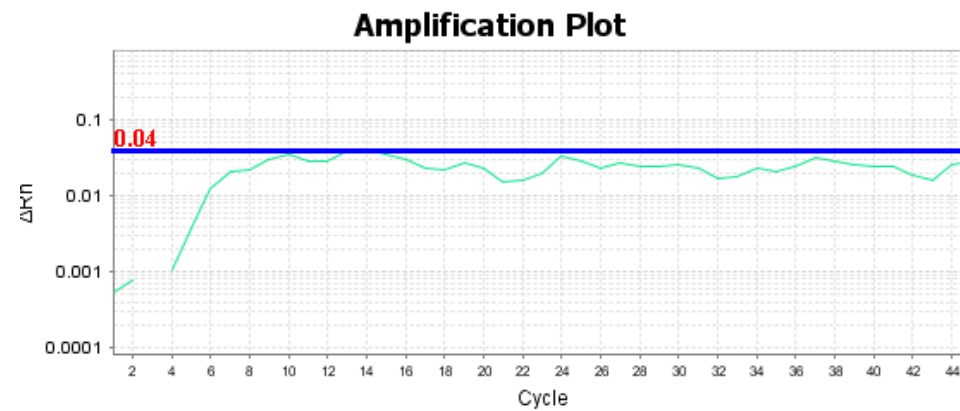
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You are not sure of your result? Use Multicomponent plot





Baseline	Baseline Start	Baseline End
AUTO	3	4



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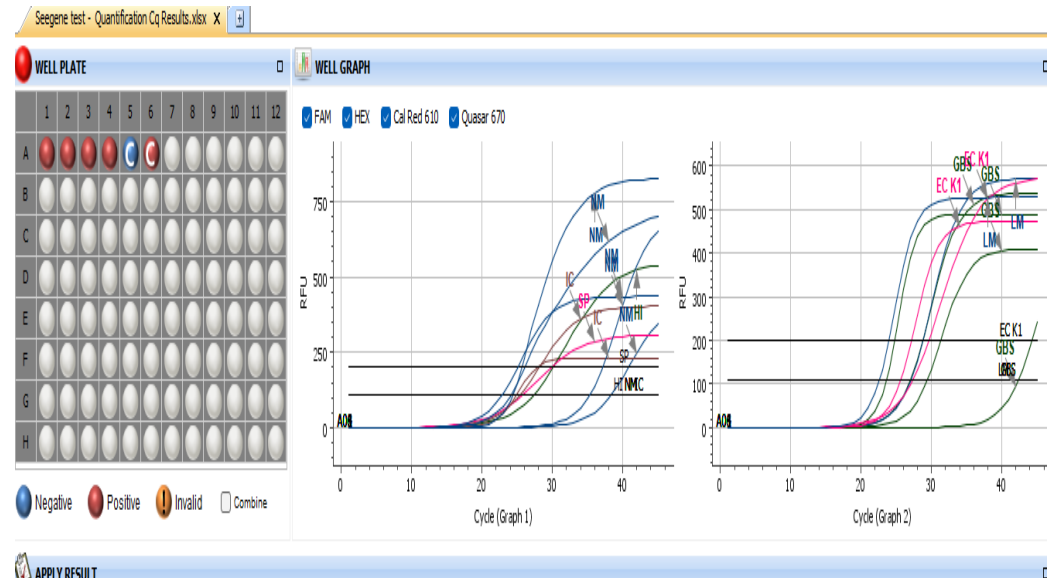


Meningitis-B Allplex Assay by Seegene

A multiplex real-time PCR assay for the detection of *Neisseria meningitidis*, *Listeria monocytogenes*, *Haemophilus influenzae*, *Streptococcus agalactiae* (GBS), *Streptococcus pneumoniae* (SP), *Escherichia coli* K1 (EC K1) from cerebrospinal fluid (CSF) specimen.



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BioFire PCR



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BioFire PCR

The Meningitis/Encephalitis Testing Menu

Discover the 14 targets on the BIOFIRE ME Panel, including seven viruses, six bacteria, and one yeast associated with community-acquired meningitis and encephalitis.

Performance

Overall 94.2% sensitivity and 99.8% specificity¹

Sample Type

Cerebrospinal Fluid (CSF) collected by lumbar puncture

VIRUSES	BACTERIA	YEAST
• Cytomegalovirus (CMV) • Enterovirus (EV) • Herpes simplex virus 1 (HSV-1) • Herpes simplex virus 2 (HSV-2)	• <i>Escherichia coli</i> K1 • <i>Haemophilus influenzae</i> • <i>Listeria monocytogenes</i> • <i>Neisseria meningitidis</i>	• <i>Cryptococcus</i> (<i>C. neoformans</i>/<i>C. gattii</i>)



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FilmArray

Meningitis / Encephalitis (ME) Panel - IVD

BIO FIRE[®]
BY BIOMERIEUX

www.BioFireDx.com

Run Summary

Sample ID: [REDACTED]

Run Date: 12 Mar 2025

3:51 AM

Controls: Passed

Detected: *Haemophilus influenzae*

Result Summary

Bacteria

Not Detected	<i>Escherichia coli</i> K1
✓ Detected	<i>Haemophilus influenzae</i>
Not Detected	<i>Listeria monocytogenes</i>
Not Detected	<i>Neisseria meningitidis</i>
Not Detected	<i>Streptococcus agalactiae</i>
Not Detected	<i>Streptococcus pneumoniae</i>

Viruses

Not Detected	Cytomegalovirus
Not Detected	Enterovirus
Not Detected	Herpes simplex virus 1
Not Detected	Herpes simplex virus 2
Not Detected	Human herpesvirus 6
Not Detected	Human parechovirus
Not Detected	Varicella zoster virus

Yeast

Not Detected	<i>Cryptococcus neoformans/gattii</i>
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Run Details



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LAMP assay

Machine



Water bath

Or



Dry bath

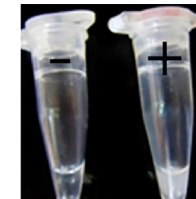
Or



Genie III instrument

Visualisation/ Detection

- Turbidity: visual inspection of the precipitate formed by magnesium pyrophosphate.
- Fluorescence: detection under UV light using intercalating dyes such as SYBR Green or EvaGreen.



Ranjbar et al.,
2016



LAMP assay

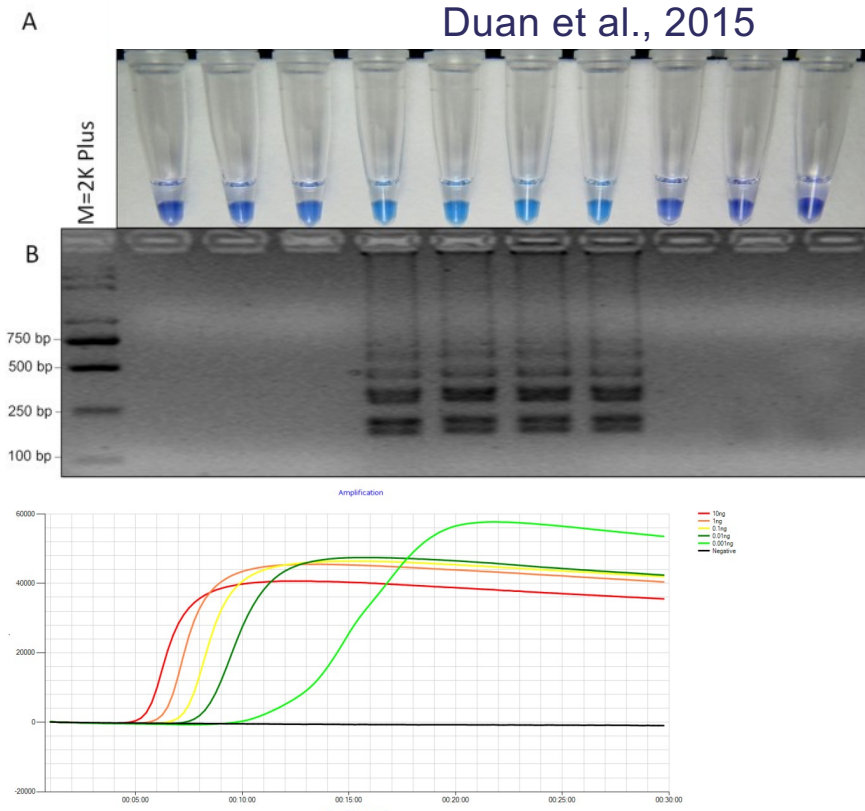
Visualisation/ detection

- Visual / Colorimetric Detection: using indicators such as Hydroxy Naphthol Blue (HNB), Calcein, Phenol Red, etc.
- Agarose Gel Electrophoresis: separation and visualization of LAMP products on a gel.
- Real-time Monitoring: amplification curves obtained on instruments such as the Genie® system.



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Thank you



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